

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1649

Desoto Sw 100 kV
October 04, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1649. The Interconnection Customer proposes a 133.9 MW solar photovoltaic facility interconnecting at the Desoto Sw 100 kV substation in the Illinois area. The request was received on January 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 446 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,783,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Cargill Wire to Desoto Sw 100 kV
Contingency: Desoto Sw to Ykk Usa Chestney 100 kV L
Loading: 105.2% (Rating: 127 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
- Facility: Ykk Usa Chestney to Desoto Sw 100 kV line
Contingency: Western Sugar Coop - Scottsbluff to Cargill Wire 100 kV L
Loading: 105.2% (Rating: 132 MVA)
DFAX: 77.8%
Pre-project loading: 26.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cargill Wire to Desoto Sw 100 kV line	Reconductor line	\$6,653,000	100% (\$6,653,000)
Ykk Usa Chestney to Desoto Sw 100 kV line	Reconductor line	\$155,000	84% (\$130,000)
	Total allocated to project		\$6,783,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.